

Pooneh Mousavi

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AI researcher and engineer finishing a PhD at Mila and Concordia University, working on **multimodal foundation models**, **audio representation learning**, and **interpretability**. My research focuses on bridging the gap between audio and large language models, from discrete audio representations to benchmarks and evaluation. Most recently worked on **generative speech models** at Epic Games. Core contributor to [SpeechBrain](#), a widely used open-source conversational AI toolkit, and founder of the [Conversational AI Reading Group](#).

Education

[Mila – Quebec AI Institute and Concordia University](#)

2022-X

PhD Candidate, Computer Science. GPA: 4.0. Advisor: Mirco Ravanelli.

Research focus: Multimodal foundation models, audio representation learning, interpretability.

[University of Texas at Dallas](#)

2018-2021

MS, Computer Science. GPA: 3.9.

[K. N. Toosi University of Technology \(KNTU\)](#)

2009-2013

BSc, Computer Engineering.

Experience

[Epic Games](#), Research Scientist Intern

8/2025-6/2026

- Generative and expressive text-to-speech models.
- Montreal, Canada. Mentor: Marc-André Carboneau.

[Mila](#), Scientist in Residence

9/2022-X

- Research on discrete audio representations, multimodal foundation models, and evaluation and interpretability of large audio language models.
- Core contributor to [SpeechBrain](#), a widely used open-source conversational AI toolkit: CommonVoice ASR recipes, dialogue systems, LLM interfaces (Llama 2, GPT), discrete audio representations, and a multilingual continual-learning ASR benchmark.

[Draft & Goal \(Mila – IVADO\)](#), ML Scientist in Residence

5/2023-8/2023

- Parameter-efficient fine-tuning of LLMs (Falcon, Llama 2) with QLoRA for domain-specific applications. Mentor: Eva Portelance.

[UKP Lab](#), TU Darmstadt, Research Scientist

10/2021-8/2022

- Applied ML and NLP to the AI in Digital Mental Health project, including analysis under severely missing multimodal data. Advisor: Iryna Gurevych.

[University of Texas at Dallas](#), Graduate Research Assistant

2018-2021

- NLP and computational social science on social media data, including hashtag-hijacking detection on Twitter. Advisor: Jessica Ouyang.

[Pioneer Natural Resources](#), Data Science Intern

1/2021-5/2021

- Real-time anomaly detection framework for electric submersible pump time-series data using LSTMs, Transformers, and autoregressive models.

Publications ([Google Scholar](#))

* Denotes equal contribution. † Denotes lead author.

- (1) **Listen First, Then Answer: Timestamp-Grounded Speech Reasoning.**
Jihoon Jeong, **Pooneh Mousavi**, Mirco Ravanelli, Cem Subakan.
Preprint, under review, 2026. [pdf](#)
- (2) **Investigating Faithfulness in Large Audio Language Models.**
Pooneh Mousavi, Lovenya Jain, Mirco Ravanelli, Cem Subakan.
Conference of the International Speech Communication Association (Interspeech), 2026. [pdf](#)
Oral Presentation
- (3) **Enhancing Audio Reasoning via Semantic Summary Prediction.**
Francesco Bonzi, **Pooneh Mousavi**, Cem Subakan, Mirco Ravanelli.
Conference of the International Speech Communication Association (Interspeech), 2026.
- (4) **DASB – Discrete Audio and Speech Benchmark.**
Pooneh Mousavi, Jarod Duret, Darius Petermann, Artem Ploujnikov, Luca Della Libera, Anastasia Kuznetsova, Cem Subakan, Mirco Ravanelli.
Transactions on Machine Learning Research (TMLR), 2026. [pdf](#)
- (5) **ALAS: An Automatic Latent Alignment Score for Audio Language Models.**
Pooneh Mousavi, Yingzhi Wang, Mirco Ravanelli, Cem Subakan.
IEEE International Workshop on Machine Learning for Signal Processing (MLSP), 2026. [pdf](#)
Also at ICML Workshop on Machine Learning for Audio, 2025.
- (6) **Discrete Audio Tokens: More Than a Survey!**
Pooneh Mousavi[†], Gallil Maimon*, Adel Moumen*, Darius Petermann*, Jiatong Shi*, Haibin Wu*, Haici Yang*, Anastasia Kuznetsova*, Artem Ploujnikov, Ricard Marxer, Bhuvana Ramabhadran, Benjamin Elizalde, Loren Lugosch, Jinyu Li, Cem Subakan, Phil Woodland, Minje Kim, Hung-yi Lee, Shinji Watanabe, Yossi Adi, Mirco Ravanelli.
Transactions on Machine Learning Research (TMLR), 2025. [pdf](#)
- (7) **LiSTEN: Learning Soft Token Embeddings for Neural Audio LLMs.**
Pooneh Mousavi*, Shubham Gupta*, Cem Subakan, Mirco Ravanelli.
Conference of the International Speech Communication Association (Interspeech), 2025. [pdf](#)
- (8) **What Are They Doing? Joint Audio-Speech Co-Reasoning.**
Yingzhi Wang, **Pooneh Mousavi**, Artem Ploujnikov, Mirco Ravanelli.
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2025. [pdf](#)
- (9) **How Should We Extract Discrete Audio Tokens from Self-Supervised Models?**
Pooneh Mousavi, Jarod Duret, Salah Zaiem, Luca Della Libera, Artem Ploujnikov, Cem Subakan, Mirco Ravanelli.
Conference of the International Speech Communication Association (Interspeech), 2024. [pdf](#)
Oral Presentation
- (10) **CL-MASR: A Continual Learning Benchmark for Multilingual ASR.**
Luca Della Libera*, **Pooneh Mousavi***, Salah Zaiem, Cem Subakan, Mirco Ravanelli.
IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP), 2024. [pdf](#)
- (II) **Open-Source Conversational AI with SpeechBrain 1.0.**
Mirco Ravanelli, Titouan Parcollet, Adel Moumen, et al.
Journal of Machine Learning Research (JMLR), 2024. [pdf](#)

(12) **WOODS: Benchmarks for Out-of-Distribution Generalization in Time Series.**

Jean-Christophe Gagnon-Audet, Kartik Ahuja, Mohammad Javad Darvishi Bayazi, **Pooneh Mousavi**, Guillaume Dumas, Irina Rish.

Transactions on Machine Learning Research (TMLR), 2023. [pdf](#)

(13) **“Please Donate for the Affected”: Supporting Emergency Managers in Finding Volunteers and Donations in Twitter Across Disasters.**

Pooneh Mousavi, Cody Buntain.

Information Systems for Crisis Response and Management (ISCRAM), 2022. [pdf](#)

(14) **Detecting Hashtag Hijacking for Hashtag Activism.**

Pooneh Mousavi, Jessica Ouyang.

EMNLP Workshop on NLP for Positive Impact, 2021. [pdf](#)

Honors and Awards

Oral presentation, Interspeech	2024, 2026
Mila travel scholarship, Women in Machine Learning (WiML)	2025
Concordia graduate travel grant, Interspeech	2024, 2025, 2026
Top 0.1% of 500,000+ participants, National Undergraduate Entrance Exam, Iran	2009

Outreach / Organizing

[Conversational AI Reading Group](#), Founder & Lead Organizer

Fall 2024-X

- Founded and run a weekly reading group and invited-talk series on conversational AI, speech, and audio foundation models, with speakers from both industry and academia.
- All sessions are recorded and made openly available on the group’s [YouTube channel](#).

Teaching

Teaching Assistant, Conversational AI, Concordia University	2023-2024
Teaching Assistant, University of Texas at Dallas	2018-2020
Teaching Assistant, K. N. Toosi University of Technology	2013

Reviewing

Conferences

International Conference on Learning Representations (ICLR)	2025
Neural Information Processing Systems (NeurIPS) , Datasets and Benchmarks Track	2024
Artificial Neural Networks in Pattern Recognition (ANNPR)	2024

Workshops

Explainable AI for Speech and Audio (XAI-SA) , ICASSP	2024
Self-supervision in Audio, Speech and Beyond (SASB) , ICASSP	2024

Skills

Software: Python (PyTorch, SpeechBrain, HuggingFace), C++, Java, Scala, JAX, TypeScript, SQL, Git, Docker, Kubernetes, Linux
Expertise: Audio processing, multimodal foundation models, discrete audio representations, LLM fine-tuning and post-training, evaluation and interpretability